# Work-energy lab

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Compare measured work with kinetic-energy change.

**SUCCESS CRITERIA - I can...**

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “trial” and “work” to compare your model or result. Sentence starters are optional.

**trial**: one run or repetition of a test

**work**: energy transferred by a force along a displacement

**displacement**: change from initial position to final position, with direction

## 01 Warm-up: bring back an earlier idea

A 3 N friction force opposes a 5 m displacement.
**Find** work, including sign and units.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and record actual trials separately.

**Worked reasoning:** Supplied measurements give net work 12 J and kinetic-energy change 11 J. Their 1 J difference could reflect uncertainty or omitted work; it is evidence to investigate.

**Your turn: Draw** two bars for these measurements and label one check of the experiment.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

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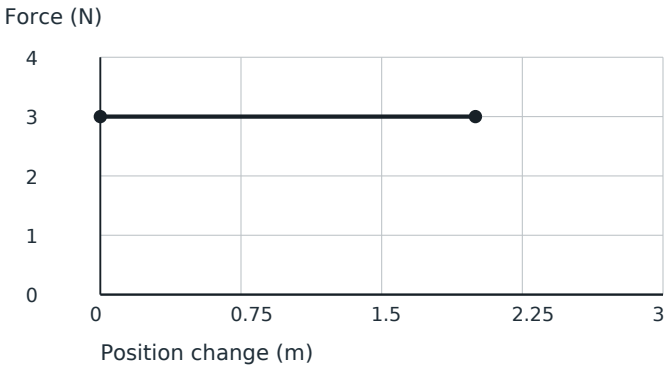
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

The supplied force-displacement graph stays at 3 N over 2 m.
**Calculate** work.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** 1 2 3

## 04 Apply the physics

A 2 kg cart slows from 3 m/s to 2 m/s during a measured interval. A force sensor reports +3 J of net work over that same interval.
**Find** initial K, final K and ΔK. Do the two measurements agree? Suggest a check.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

Create a trial table with mass, initial/final speed, work and ΔK.
**Name** two causes of disagreement that deserve checking.

**Sentence starter:** I notice ... . This supports ... because ... .

**Marzano check - circle:** 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 06 Review: connect an earlier idea

Two equal 4 kg carts approach at 6 m/s in opposite directions and stick.
**Find** their common final velocity and explain.
**Use** before and after snapshots to organize the momentum account.

**Word help:** final = at the end of the interval; subscript f

**Sentence starter:** My claim is … . The diagram/data show …, so … .

| Before | After |
| --- | --- |
| | |

Label mass, velocity and momentum.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Why must the work and speed change refer to the same start and end positions?
**Use** the energy model to show the quantities you compare; label units and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

| Initial | Transfers | Final |
| --- | --- | --- |
| | | |

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Two trials have differences 1 J and 5 J, each with possible difference uncertainty ±2 J. Which is consistent with zero under those bounds?
**Use** the energy model to show the quantities you compare; label units and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

| Initial | Transfers | Final |
| --- | --- | --- |
| | | |

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3